



"Together, No Stock-Out."

AI-powered healthcare supply intelligence for India's Primary Health Centres

 Google Gemini AI

 72 Demo PHCs

 3 States

 Synthetic Demo Data

Build with AI: Code for Communities — Second Edition | Hack2Skill | Team SahaMatrix

THE PROBLEM

India's PHC Supply Chain is Broken



Medicine Stock-Outs

Life-saving drugs — insulin, ORS, antimalarials — run out during seasonal surges while nearby PHCs hold expiring surplus.



Delayed Detection

No real-time cross-facility visibility. By the time a stock-out is reported, patients are already denied medicines.



Paper-Based Inventory

Rural PHCs still rely on handwritten stock registers. Manual reporting creates days of lag before critical shortages surface.



No Peer Redistribution

No mechanism to identify surplus at adjacent facilities and coordinate emergency transfers without waiting for central directives.

THE SOLUTION

SahaMatrix in One Diagram

Data → Prediction → AI → Decision → Action → Impact



End-to-End Operational Flow

1 PHC Stock Data Ingested
72 simulated PHCs across Maharashtra, Uttar Pradesh, Tamil Nadu with synthetic daily consumption data. Architecture supports CSV upload and paper register OCR for future real-data onboarding.

2 14-Day Demand Forecast
Per-facility Ridge regression model calculates daily consumption trend. Forecasts days-of-stock remaining per medicine. Recharts visualization displayed per PHC.

3 Stock-Out Alerts Ranked
Critical (<3 days) and Deficit (<7 days) alerts surfaced in ranked queue. Severity badges, daily consumption rate, and stock runway shown per facility.

4 Gemini AI Explains the Alert
One click calls Gemini Flash with grounded telemetry context. Returns: root cause, stock runway analysis, and recommended intervention. Explicitly labeled "Powered by Google Gemini".

5 AI Redistribution Recommended
Greedy Haversine proximity algorithm identifies surplus PHC nearest to deficit PHC. Intra-state transfers prioritized; inter-state orchestrated when necessary. Transfer cards shown in Transfers tab.

6 Transfer Applied, Dashboard Updates
Single or batch apply triggers REST API call. MySQL stock records updated. Map transfer polylines animate. KPI "Stock-outs Prevented" counter increments live.

PLATFORM CAPABILITIES

Key Features

INTELLIGENCE	GEMINI AI	INTERACTION	OPERATIONS
 Stock-Out Prediction Ridge regression forecasts 14-day demand curves per PHC per medicine. Days-of-stock runway computed continuously.	 Grounded Alert Insights Gemini Flash explains each stockout alert with root cause, clinical urgency, and recommended action — grounded in live telemetry only.	 Voice AI Q&A Web Speech API microphone dictation. Ask natural-language questions about active alerts. Gemini answers; speech synthesis reads aloud in selected language.	 Peer Redistribution Greedy Haversine algorithm identifies nearest surplus PHC. Single and batch apply transfers. Map polylines animate the route. MySQL stock updated.
 Fed. Learning Simulation FedAvg simulation across 3 state nodes. MAE bar charts and round-loss curve. Privacy: only model weights aggregated, not raw data.	 Situation Briefings 5-bullet executive brief for state health directors, synthesized by Gemini from live supply data. One-click copy for WhatsApp or memos.	 Register OCR Gemini multimodal vision extracts medicine names and quantities from photos of handwritten paper stock registers. Synonym normalization (PCM → Paracetamol).	 Multilingual UI English, Hindi, Marathi, Tamil runtime translation via LanguageContext. All UI labels, alerts, and AI responses in selected language.

Gemini Flash Integration

Four distinct Gemini AI endpoints — all strictly grounded in live supply telemetry, zero patient identifiers



Clinical Alert Explanation

POST /api/ai/explain-alert

- ✓ Grounded root-cause analysis for each stock-out using live PHC telemetry
- ✓ Clinical urgency scoring and recommended intervention steps
- ✓ Responds in the user's selected language (EN/HI/MR/TA)
- ✓ "Powered by Google Gemini" badge vs "Fallback mode" label shown transparently



Executive Situation Brief

POST /api/ai/situation-brief

- ✓ 5-bullet summary for state health leadership synthesized from supply metrics
- ✓ Per-state scoping (MH / UP / TN) with per-state drug risk profiling
- ✓ One-click copy for WhatsApp or district officer memo distribution
- ✓ Text-to-speech read-aloud via Web Speech API



Voice-First Q&A Assistant

POST /api/ai/ask

- ✓ Natural-language questions over active alerts — spoken or typed
- ✓ Web Speech API microphone dictation with browser-compatibility fallback
- ✓ Grounded answers: "Use only the supplied telemetry. Do not invent facts."



Paper Register OCR

POST /api/ai/ingest-register

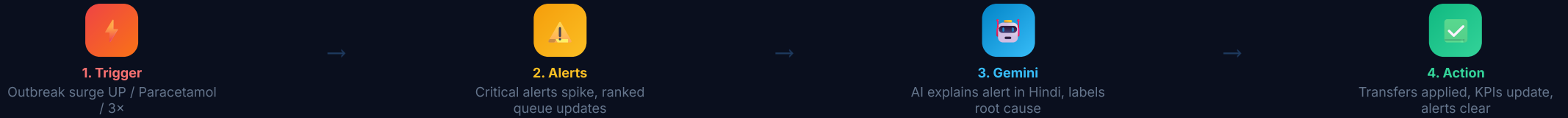
- ✓ Gemini multimodal vision reads photos of handwritten/printed stock registers
- ✓ Synonym normalization: "PCM" → Paracetamol, "ACT" → Artemisinin ACT
- ✓ Confidence scoring and editable review table before committing to stock



Gemini API key stored securely on the Spring Boot backend only — never exposed to the browser or frontend code.

LIVE DEMO STORY

3-Minute End-to-End Walkthrough



Architecture Stack

LAYERED ARCHITECTURE



What Is Real vs Demo Data

✓ Real-World Context

- ✓ **Geographic data:** Actual Indian state and district names — Maharashtra (Nagpur, Pune), Uttar Pradesh (Lucknow, Kanpur, Varanasi), Tamil Nadu (Chennai, Coimbatore, Madurai).
- ✓ **Medicine list:** Clinically valid medicines — Paracetamol, Amoxicillin, ORS, Insulin, Artemisinin ACT, Iron-Folic Acid, Metformin, Cotrimoxazole, Measles Vaccine.
- ✓ **Problem framing:** Based on documented challenges at India's PHC network (~30,000 facilities nationally).
- ✓ **Algorithm logic:** Haversine geodesic distance, FedAvg aggregation protocol, Ridge regression forecasting — all standard, validated approaches.
- ✓ **AI responses:** Strictly grounded in live telemetry provided to Gemini; zero hallucinated patient data.

🧠 Synthetic Demo Data

- ⚠️ **PHC stock levels:** Synthetically generated using seed-42 reproducible data. Not sourced from real government or HMIS systems.
- ⚠️ **Consumption figures:** Simulated daily consumption rates. Not actual patient dispensing records from any PHC.
- ⚠️ **PHC locations:** Representative lat/lng coordinates within correct districts, not verified GPS positions.
- ⚠️ **Federated learning:** Simulated FedAvg with synthetic state-node separation. Not a live distributed training deployment.
- ⚠️ **User uploaded images:** Test register images supplied for OCR demo; architecture supports real uploaded images.

🏗️ Real-Data Path (Architecture-Ready)

The backend DataStore, JPA entity layer, and CSV import modal are designed to ingest real PHC data from e-Aushadhi, HMIS CSV exports, or direct API feeds — with zero architectural changes required.

MEASURABLE OUTCOMES

What SahaMatrix Demonstrates

Based on actual prototype behavior — no invented metrics



Earlier Stock-Out Detection

14-day demand forecasting surfaces stock-out risk days before they occur — vs paper-based systems where the shortage is only discovered when the shelf is empty.



AI-Assisted Decision Support

Gemini provides grounded clinical context for each alert, replacing manual district-officer analysis. Executive situation briefs generated in seconds vs hours of manual report compilation.



Cross-PHC Visibility

Unified GIS dashboard shows risk status of all 72 demo PHCs simultaneously. Decision-makers can compare facilities across districts and states in one view — without phone calls or paper reports.



Coordinated Redistribution

Greedy Haversine routing identifies the nearest viable surplus PHC. Batch redistribution applied with one click. Full transfer audit trail maintained in MySQL.

72

Demo PHCs
monitored

14

Day demand
forecast horizon

4

Gemini AI
integrations

4

Languages
supported

FUTURE SCOPE

Roadmap to Production

All items below are clearly labeled as **future scope** — not present in the current prototype

DATA & INTEGRATION



Real Government PHC Data: Integration with e-Aushadhi, HMIS, and NHM data pipelines. Direct API feeds from state health department systems replacing synthetic demo data.



Live Data Feeds: Real-time stock consumption pushed from point-of-care dispensing systems — eliminating manual register entry and CSV upload dependency.



Broader Health Resource Monitoring: Extend beyond medicines — beds, oxygen, vaccines, diagnostic equipment, and human resources across facilities.

PLATFORM & SCALE



Production-Grade Auth & Security: OAuth2 / SSO integration with NIC login. Role-based access control enforced at API layer. Audit logging for all stock mutations.



Real Distributed Federated Learning: Current prototype simulates FedAvg in a single JVM. Production would run actual state-separated compute nodes with encrypted weight transmission.



Mobile-First Field App: Progressive web app or native Android app for PHC pharmacists to submit daily stock counts, receive alerts, and request emergency resupply — even with intermittent connectivity.



National Scale Deployment: Cloud-native architecture designed for 30,000 PHCs across all 28 states. Performance benchmark prototype demonstrated <6s for 5,000-facility redistribution on single machine.



Together, No Stock-Out.

SahaMatrix — AI-powered healthcare supply intelligence, built with Google Gemini, for India's 30,000 Primary Health Centres.

SOURCE CODE

github.com/The-kanak/SahaMatrix

Full stack: Spring Boot + React + MySQL + Gemini

TECH STACK

Google Gemini

Spring Boot

React 18

MySQL

Leaflet

Docker

 Powered by Google Gemini Flash

 Demo uses synthetic PHC data

 20/20 smoke tests passing